

Project Overview

We trained and evaluated various **Sentence Embedding** architectures based on transformer encoders, specifically focusing on the interaction between different **Pooling Methods** (Mean, Max, Attention) and **training objectives**. We benchmark these models on standard datasets to determine optimal configurations for semantic similarity tasks.

Pooling Strategies

We investigate three primary aggregation strategies applied over the token embeddings $H = [h_1, \dots, h_n]$ output by BERT.

- **Mean Pooling:** Average of all token embeddings.
- **Max Pooling:** Maximum value in every hidden dimension.
- **Attention Pooling:** Learnable attention pooling over token embeddings

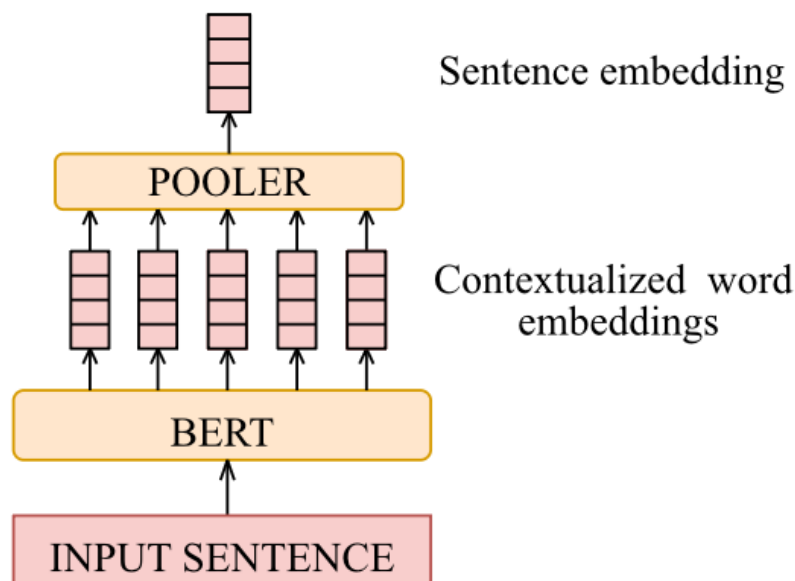


Fig 1: Token embeddings passing through pooling.

Attention Mechanism

Unlike static pooling, attention pooling learns to weight specific tokens (embeddings) higher than others. We evaluated attention pooling on the final layer versus multi-layer attention pooling.

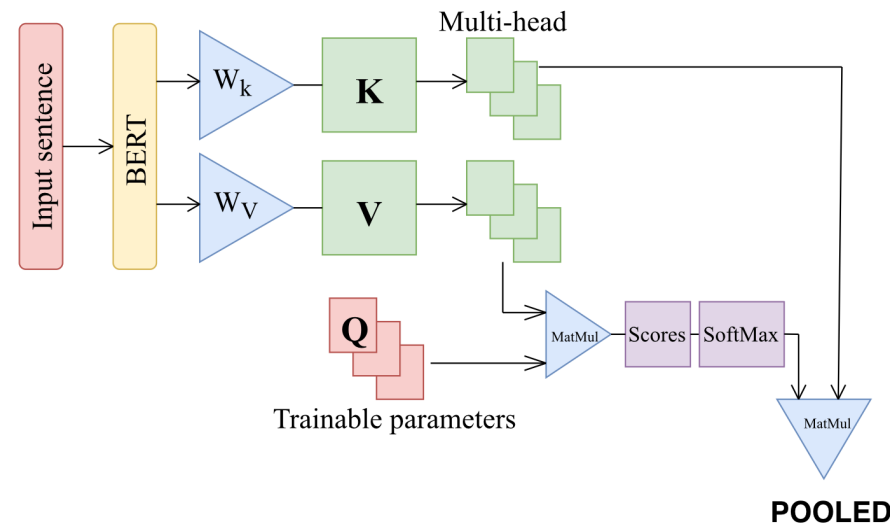


Fig 2: Detailed view of the Attention Pooling mechanism.

Loss Objectives

We optimize the vector space so that semantically similar sentences are close.

Loss Functions

$$\mathcal{L}_{triplet} = \max(\|s_a - s_p\| - \|s_a - s_n\| + \epsilon, 0)$$

$$\mathcal{L}_{mnr} = -\log \frac{e^{\text{sim}(s_a, s_p)/\tau}}{\sum_{j \in \text{batch}} e^{\text{sim}(s_a, s_j)/\tau}}$$

$$\mathcal{L}_{class} = \text{CE}(\text{softmax}(W_t[u; v; |u - v|]), y)$$

- **Triplet Loss:** Makes the distance between the **anchor** and **positive** smaller than the distance to the **negative**.
- **MNR (Multiple Negatives Ranking):** Uses in-batch samples as negatives for higher efficiency.
- **SBERT Classification:** Concatenates embeddings and predicts NLI labels (Entailment, Neutral, Contradiction)

Experimental Setup

Data: Trained on SNLI and MNLI datasets (500k+ pairs).

Backbone: bert-base-uncased (110M params).

Eval: Evaluated on STS Benchmark (Spearman correlation).

Results

Performance evaluation on the STS Benchmark (Spearman's rank correlation $\rho \times 100$).

Model Configuration	STS ρ
1. Impact of Loss Function (Mean Pooling)	
Pair-Class Classification (Softmax)	75.6
Triplet Loss	77.4
MNR Loss (Batch Size = 32)	80.9
MNR Loss (Batch Size = 64)	82.0
2. Impact of Pooling Strategy (MNR Loss, batch size 32)	
Max Pooling	78.8
Mean Pooling	80.9
Last-Layer Attention	79.6
Multilayer Attention	80.2
3. Baselines & Reference	
BERT-base (mean pooling, no fine-tuning)	46.3
SBERT-base (Reimers et al., 2019)	77.3
SimCSE-BERT-base (Gao et al., 2021)	84.2

Key Insight

- **Pooling:** Simple *Mean Pooling* proved robust and efficient, matching or outperforming attention mechanisms for this task.
- **Batch Size Matters:** MNR Loss benefits significantly from larger batches ($B = 64$) as it increases the number of negative samples per anchor.